THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET B

Chapters 11 & 12: **Exploring Some Geometric Themes, Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A figure which has the same shape at every scale, repeating itself, is called a: 1

(a) polygon **(b)** fractal **(c)** net **(d)** prism

2. The number of squares in the net of a cube is: 1

(a) 12 **(b)** 6 **(c)** 4 **(d)** 8

3. A pyramid with a square base has triangular faces numbering: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

4. The side view of a cone is a: 1

(a) triangle **(b)** square **(c)** rectangle **(d)** circle

5. The relation between faces, vertices and edges of a polyhedron is: 1

(a) $F + V - E = 2$ **(b)** $F + E - V = 0$ **(c)** $F + V + E = 2$ **(d)** $F - V + E = 2$

6. A line graph is most useful for showing: 1

(a) change over time **(b)** colours **(c)** a single value **(d)** parts of a whole

7. The difference between the highest and lowest values is called the: 1

(a) mode **(b)** median **(c)** range **(d)** mean

8. For the data 2, 2, 3, 4, 9, the best measure of a typical value unaffected by 9 is the: 1

(a) range **(b)** sum **(c)** median **(d)** mean

9. The data 10, 10, 10, 10 has a mean of: 1

(a) 1 **(b)** 10 **(c)** 40 **(d)** 0

10. If the mean of 5 numbers is 8, the sum of the numbers is: 1

(a) 8 **(b)** 40 **(c)** 1.6 **(d)** 13

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Name the cross-section when a cylinder is cut parallel to its base. 2

12. A cube has edge 4 cm. Find the total length of all its edges. 2

13. Is the mean always one of the data values? Give an example. 2

14. How does adding 10 to every observation change the median? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A cube has edge 6 cm. Find its surface area and the number of such cubes needed to make a cuboid of dimensions $12 \times 6 \times 6$. **3**

16. A line graph shows sales: Jan 20, Feb 25, Mar 30, Apr 28, May 35. Find the month of greatest rise and the average sales. **3**

17. Draw the nets of a cube and a cuboid. Mention one difference. **3**

OR

Describe the Koch snowflake construction in three steps. **3**

18. The mean weight of 8 boys is 45 kg. When one boy leaves, the mean becomes 44 kg. Find the weight of the boy who left. **3**

OR

Draw a dot plot for 2, 3, 3, 4, 4, 4, 5 and state the most frequent value. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Why do prisms and pyramids differ in nets?
(ii) Draw the net of a triangular pyramid in words. **5**

OR

Case study: Packaging for a chocolate is a triangular prism with equilateral triangle base of side 4 cm and length 20 cm. **5**
(i) Number of faces, edges, vertices.
(ii) Sum of lengths of all edges.
(iii) Shape of the front view.

20. **Case study:** Marks of two students in 5 tests: A: 70, 72, 74, 76, 78; B: 40, 60, 74, 88, 98. **5**
(i) Find both means.
(ii) Which is more consistent?
(iii) Find the ranges.

OR

(i) Find x if the median of 3, 5, x, 9, 11 is 8. **5**
(ii) Find the mean.

END OF QUESTION PAPER — Total: 40 Marks

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